

Linear Functions & Inequalities



Slope and slope-intercept form

- 1 Find the slope of the line passing through $(2, 3)$ and $(6, 11)$.
 - 2 Write the equation of the line with slope -3 passing through $(1, 4)$, in slope-intercept form.
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Parallel and perpendicular lines

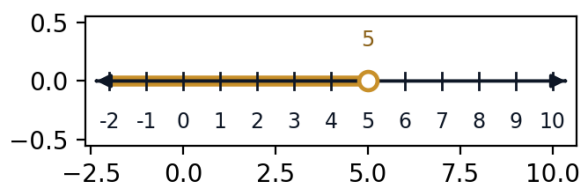
- 3 Find the equation of the line parallel to $y = 2x - 5$ passing through $(3, 4)$.
 - 4 Find the equation of the line perpendicular to $y = \frac{1}{2}x + 3$ passing through $(4, 1)$.
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Finding equations from two points or a table

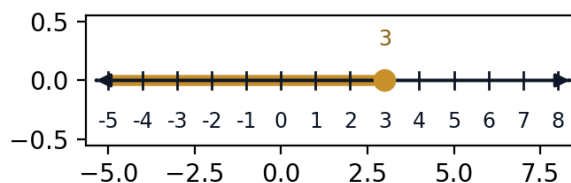
- 5 A linear function passes through $(0, 5)$ and $(4, 17)$. Find its equation.
 - 6 A table shows $x = 1, y = 7$ and $x = 3, y = 15$ for a linear function. Find the equation.
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Solving and graphing linear inequalities

- 7 Solve the inequality $3x - 4 < 11$, and graph the solution on a number line description.

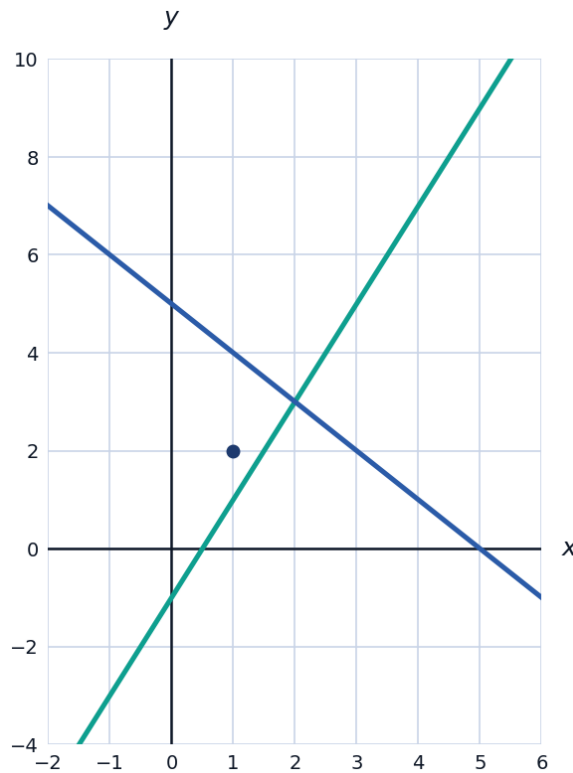


- 8 Solve the inequality $-2x + 7 \geq 1$, being careful with the direction of the inequality sign.



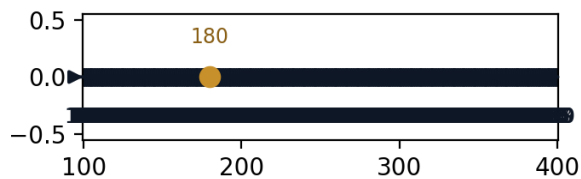
Systems of linear inequalities

- 9 A system requires $y > 2x - 1$ and $y \leq -x + 5$. Determine whether the point $(1, 2)$ satisfies both inequalities.



Interpreting linear inequalities in context

- 10 A student has 50 dollars and wants to buy notebooks costing 6 dollars each, keeping at least 8 dollars in reserve. Write an inequality for the number of notebooks n the student can buy, and solve it.
- 11 A factory produces at least 200 units per day but at most 350 units per day. Write a compound inequality for the daily production p , and state whether $p = 180$ is a valid production level.

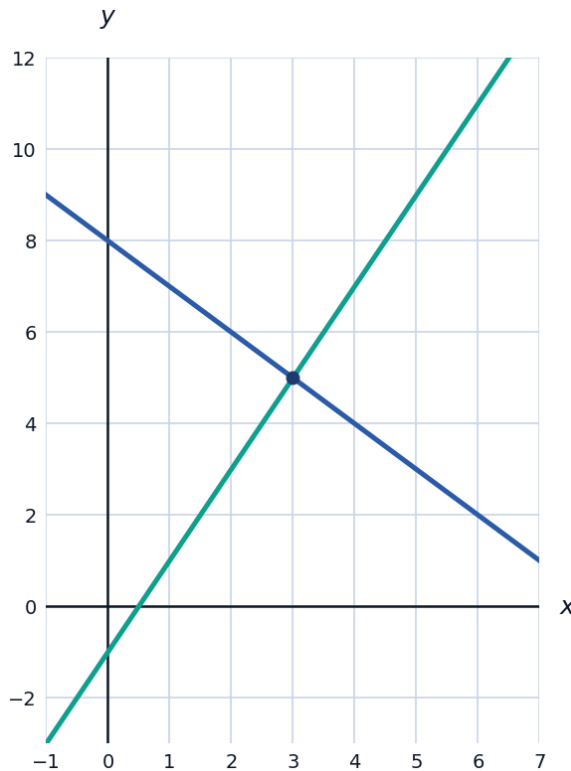


Comparing rates of change between two linear functions

- 12 Function f is given by $f(x) = 3x + 2$. Function g is defined by the table: $x = 0, g = 1$ and $x = 2, g = 9$. Which function has the greater rate of change?

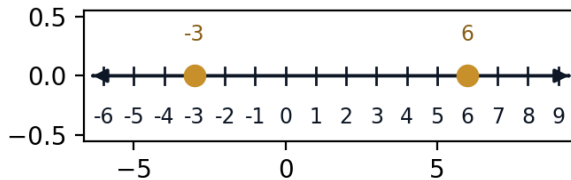
Finding the intersection of two lines graphically and algebraically

- 13** Two lines are given by $y = 2x - 1$ and $y = -x + 8$. Find their point of intersection, and explain what this point represents graphically.

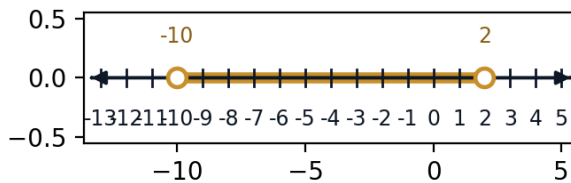


Absolute value equations and inequalities

- 14** Solve $|2x - 3| = 9$.



- 15** Solve $|x + 4| < 6$, giving your answer as an interval.

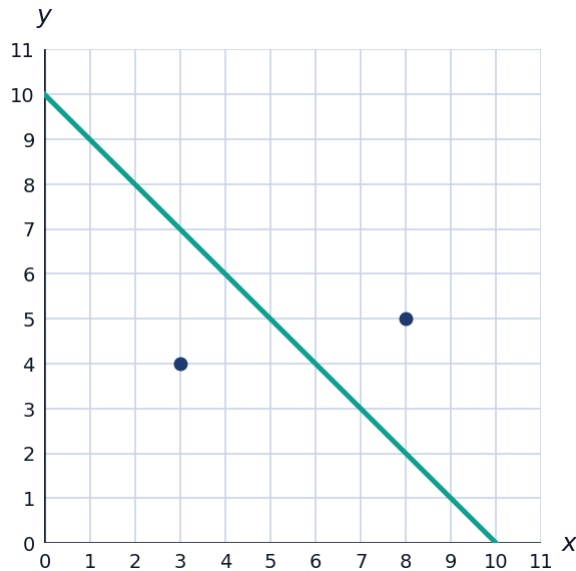


Writing a linear inequality from a verbal description

- 16** A shipping company charges 3 dollars per pound but offers free shipping if the package weighs less than 2 pounds. Write a piecewise description of the cost $C(w)$ as a function of weight w , and find the cost of a 5-pound package.
- 17** A parking garage charges 5 dollars for the first hour and 2 dollars for each additional hour. Write an equation for the total cost C after h hours (for $h \geq 1$), and find the cost for 6 hours.
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Graphing a system of inequalities and identifying the feasible region

- 18** A system requires $x + y \leq 10$ and $x \geq 2$ and $y \geq 0$. Determine which of the points $(3, 4)$ and $(8, 5)$ lie in the feasible (solution) region, checking each condition.



Rate of change from a real-world linear model

- 19 A water tank starts with 500 gallons and drains at a constant rate. After 4 hours, it contains 340 gallons. Write a linear equation for the volume V after t hours, and predict when the tank will be empty.

